

Nafees S

✉ nafees.s2005@gmail.com ☎ +91-99728 47009 in nafees-s 🌐 lucenity0

Education

B.M.S. College of Engineering — *B.E. in Computer Science & Engineering* CGPA: *Expected June 2027*
8.21

Narayana PU College — *12th, Karnataka State Board* **94.33%** *May 2023*

Technical Skills

Languages: Python, Java, C, JavaScript, TypeScript, Swift, SQL

Frameworks & Databases: FastAPI, React, Tailwind CSS, SwiftUI, PostgreSQL, SQLAlchemy, Alembic, Redis, Celery, Vite

AI / ML: LLMs, Prompt Engineering, Gemini API, Reinforcement Learning (PPO), Deep Learning, NLP, Multimodal Fusion, CLIP, LangChain, ChromaDB

Tools: Git/GitHub, Docker, AWS EC2, Xcode, Figma, Jupyter Notebook, VS Code

Coursework: Machine Learning, Linear Algebra, Probability & Statistics, Data Structures & Algorithms

Projects & Research

Askcal — **AI-Powered Daily Scheduler** — FastAPI, Gemini, PostgreSQL, SwiftUI,  [Source Code](#) 
Google OAuth

- Built a regret-ranked inbox classifier using Gemini structured output — extracts signals from Gmail and computes a deterministic 0–100 regret score per email, auto-converting actionable mail into tasks slotted around real Google Calendar events
- Shipped full monochrome iOS app (SwiftUI) with Today, Inbox, Calendar, Routines, and Review views; 79 backend tests passing; async SQLAlchemy + JWT auth with opaque refresh token rotation

Interpretable PPO-Based Adaptive Traffic Signal Control — Python, SUMO,  [Source Code](#) 
TraCI, RL

- Implemented a PPO agent with curriculum learning on increasing traffic complexity; integrated attention-based feature attribution to address the black-box challenge in AI-driven infrastructure
- Engineered real-time bidirectional SUMO–TraCI communication (IoT-style feedback loop) with constrained emergency preemption and priority queuing

Tiket — **Full-Stack Ticket Booking System** — SwiftUI, FastAPI, PostgreSQL,  [Source Code](#) 
AWS EC2, JWT

- Designed and deployed a full-stack ticket booking system with concurrency-safe seat allocation (PostgreSQL `SELECT FOR UPDATE`), JWT auth, and AWS EC2 deployment — load tested at 12,441 concurrent requests with zero double bookings
- Built production-grade iOS frontend in SwiftUI with real-time seat selection and dynamic barcode generation

Schedulr — **OS Concepts Simulator** — React, TypeScript, Vite, Tailwind CSS  [Source Code](#) 

- Implemented 12+ classical OS algorithms (CPU scheduling, page replacement, disk scheduling) in a fully type-safe, responsive SPA with real-time Gantt chart visualization — actively used by junior students at BMSCE

Leadership

Senior Coordinator, Phase Shift 2025 — *BMSCE* *2024 – 2025*

- Co-led logistics for BMSCE's annual technical fest; previously Junior Coordinator for Utsav 2025, managing design team and on-ground event operations